

EXERCISES AND EXACT SOLUTIONS

Quality Is a Decision Variable

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C03
Notebook	V7_0_N03
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add a subgroup completeness check.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Explain why averaging can be unsafe.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Propose a revision policy.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Distinguish concept drift from data drift.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Group by the policy-relevant subgroup and compare both unweighted and weighted missingness with disclosure controls.

Solution 2

A high score can mask a failed key, unlawful source, or unusable delay; critical gates must override averages.

Solution 3

Preserve vintages, reasons, approval, release status, notification, and reproducible recalculation.

Solution 4

Data drift changes input distributions; concept drift changes the relationship between inputs and the target/outcome.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated